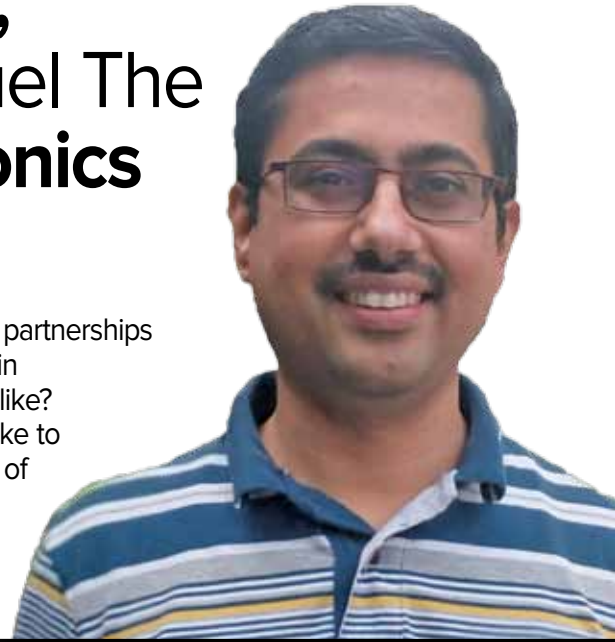


“We Aim To Boost Skills, Spark Innovation, And Fuel The Growth Of India’s Electronics Manufacturing Sector”

Everyone advises industry and academia to collaborate, but such partnerships are rare. IIT Kanpur and NMTronics have recently partnered to train professionals in electronics manufacturing. What will this lab look like? What are its goals and plans? Electronics For You’s Akanksha spoke to Prof. Nilesh Badwe, Professor-In-Charge of the NMTronics Centre of Excellence for Electronics Manufacturing and Skills Development, IIT Kanpur, and here are the key extracts from their interaction

PROF. NILESH BADWE
IIT KANPUR



What are this lab’s goals, and how does partnering with industry amplify its impact?

Bridging India’s electronics manufacturing skills gap is our mission! We focus on three core areas: delivering in-depth training for technicians, advanced training for R&D engineers, and executive training for top management. We are also pioneering research in material science for electronics, tackling issues like PCB warpage, and enabling prototype development for startups and large manufacturers. Powered by NMTronics’ CSR initiative, we aim to boost skills, spark innovation, and fuel the growth of India’s electronics manufacturing sector.

Is this centre a crossroad for industry experts and students alike?

Our centre is a hub for industry professionals and students, dedicated to creating a collaborative space for learning and innovation. With guidance from both technical and administrative boards, we

will fine-tune our curriculum and operations to cater to those eager to grow in India’s electronics manufacturing sector. Open to all, we strive to nurture a vibrant ecosystem for advancement and innovation in the field.

How will industry experts be involved?

Industry experts will be involved in training and knowledge dissemination. These professionals will either directly teach or train designated trainers to pass on the knowledge to the students and other users. This will ensure that the curriculum remains relevant and current with industry trends and practices. We aim to bring in leading industrial experts, including those from NMTronics, Intel, material suppliers like MacDermid Alpha, and many more, to contribute to the centre’s activities.

Can you share some details about this lab’s equipment?

Our lab will feature cutting-

edge equipment essential for a fully automated SMT line, including solder paste printers, inspection systems, pick-and-place machines, reflow ovens, and automated optical inspection systems. These tools are vital for training and research in advanced electronics manufacturing processes.

How will the curriculum adapt to industry needs and advancements?

Our flexible SMT curriculum will cater to all expertise levels, with three tiers covering basic equipment training to advanced technical skills. We shall ensure relevance by establishing a technical board for ongoing curriculum evolution based on industry feedback, providing a tailored learning experience. The programme includes theoretical, visual, and hands-on components for a comprehensive educational journey. While the training programme is in the development stage, we welcome inputs for adjustments to meet industry needs. Our training targets technicians, engineers, and high-level management, aiming to